Problem Statement

In computer science, a set is an abstract data type that can store certain values, without any particular order, and no repeated values(Wikipedia). is an example of a set, but is not a set. Today you will learn how to use sets in java by solving this problem.

You are given pairs of strings. Two pairs and are identical if and . That also implies is *not* same as . After taking each pair as input, you need to print number of unique pairs you currently have.

Complete the code in the editor to solve this problem.

Input Format

In the first line, there will be an integer denoting number of pairs. Each of the next lines will contain two strings seperated by a single space.

Constraints:

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- Length of each string is atmost and will consist lower case letters only.

Output Format

Print lines. In the line, print number of unique pairs you have after taking pair as input.

Program:

```
import java.util.*;


public class Solution {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);


    int n = sc.nextInt(); // number of pairs
    sc.nextLine(); // consume newline


    Set<String> uniquePairs = new HashSet<>();


    for (int i = 0; i < n; i++) {
      String first = sc.next();
      String second = sc.next();


      // Create a combined string for the pair
```

```
            String pair = first + " " + second;

            // Add to set (duplicates automatically ignored)
            uniquePairs.add(pair);

            // Print current number of unique pairs
            System.out.println(uniquePairs.size());
        }

        sc.close();
    }
}
```

Sample Input

```
5
john tom
john mary
john tom
mary anna
mary anna
```

Sample Output

```
1
2
2
3
3
```